

Load-Aware A* Path Planning for Quadruped Robots

Day 15 of 30 · Week 2: Refinement & Integration

NAME : HARINI P (Department of Computer science and Engineering)

1. Nature of Work

Day 15 was the Week 2 checkpoint -- the Methodology section needed to be polished, complete, and presentation-ready. After finishing the four core writing subsections across Days 11-14 (Problem Formulation, Standard A*, Load-Aware A*, and Implementation Details), today's work focused on quality: reviewing every equation for consistency with the code, adding two figures that were referenced but not yet drawn, proofreading for grammar and flow, and beginning to revise based on feedback from Member 4 (who reviewed the Problem Formulation and Standard A* subsections). The revision process also addressed one structural issue flagged by Member 4: the transition from Standard A* to Load-Aware A* was abrupt -- the reader could follow both algorithms independently but could not clearly see why the modification was necessary. A short motivating paragraph was added as a bridge between the two subsections.

2. Tasks Completed

- **Review Methodology**

All four subsections reviewed against the code output. 16 equations cross-checked with actual computation. All notation verified consistent with the Day 11 notation table

3. REVISION MADE -- BRIDGE PARAGRAPH ADDED

"While Standard A* produces the geometrically optimal path on a given grid, it has no awareness of the physical conditions the robot encounters along that path. When a quadruped carries a payload, the distribution of mass shifts the Centre of Mass away from the centroid of the support polygon -- the robot becomes progressively less stable as the payload moves off-centre. A path that is geometrically short may be mechanically unsafe if it routes the robot through cells where this shift is large. The Load-Aware A* algorithm, introduced in the following subsection, addresses this limitation by augmenting the Standard A* cost function with a continuous stability penalty that quantifies this risk at every cell."

4.METHODOLOGY -- FINAL STATUS

Section	Day Completed	Written Length	Equations	Figures	Status

A. Problem Formulation	Day 11 ▾	~550 words	Eq. 1-8	8 ▾	Final ▾
B. Quadruped Model	Day 11 ▾	~350 words	Eq. 3-6	6 ▾	Final ▾
C. Standard A*	Day 12 ▾	~700 words	Eq. 9-12	5 ▾	Final ▾
D. Load-Aware A*	Day 13 ▾	~750 words	Eq. 13-16	5 ▾	Final ▾
E. Implementation Details	Day 14 ▾	~500 words	Eq. 17	1 ▾	Final ▾
F. System Architecture	Day 14 ▾	-	-	Fig. 2+3 ▾	Final ▾
TOTAL	- ▾	~2,850 words	Eq. 1-17	17 ▾	Ready ▾